CMG PES Neuromorphic Residual Control Research Brief

Current evidence hardware hypothesis and publication strategy | 23 September 2026

Current conclusion

The project now supports a credible control result: a bounded online CMG-PES residual improves a strong quaternion PD controller with desired-rate feedforward on four smooth attitude missions across two MuJoCo multirotor models. The saturation-aware output mask prevents the learned correction from opposing a baseline axis already at its torque limit. It reduces, but does not reverse, the remaining Hardstep loss. The evidence therefore supports an ACC 2027 submission while preserving a clear hardware-validation path.

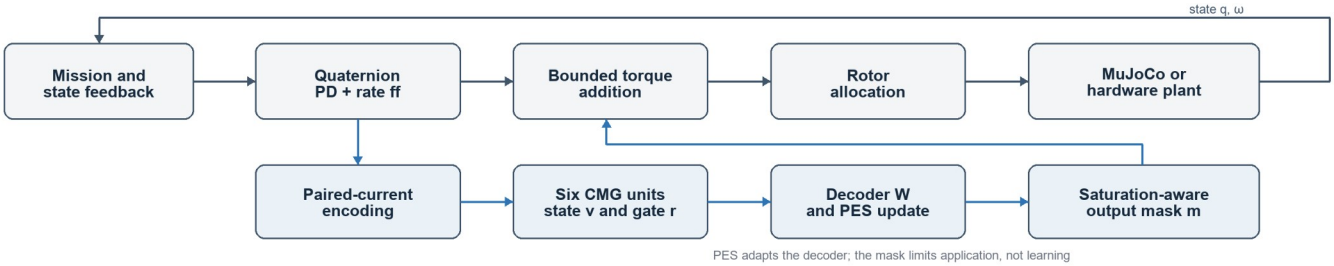


Figure 1 Control flow. The learned term is bounded before the unchanged allocation and plant interfaces; PES updates the decoder online.

What is being tested

Baseline. The tuned controller produces the saturated request *u0*. Residual path. Paired currents drive six continuous membrane-gate units with states *v* and *r*; the activity feeds an 18-weight decoder. PES adapts those weights from quaternion and rate error. The applied correction is $\delta = \beta \ell \odot m \odot s$, where β sets authority, ℓ is the torque bound, s is the clipped decoder output, and m is the saturation-aware mask. Thus $|\delta i| \le \beta \ell i$ and $\beta = 0$ recovers the baseline exactly.

Current numerical evidence

Evidence	X2	CF2	Interpretation
Five-mission geometric mean	37.3%	22.4%	Gated CMG-PES vs tuned PD+ff at ρ = 3
Four smooth missions	12.4–78.4%	5.4–59.9%	Roll, Full3, Hard45, and Yaw360 all improve
Hardstep	−2.0%	−0.2%	Near parity; the residual does not beat the baseline
Repeated torque bursts	14.4%	5.9%	Reduction in burst-phase error during Hover
Specified sensor noise	11.3 / 10.8%	3.9 / 4.0%	Full3 / Hard45 at the higher tested noise level

Metric: mean shortest-path quaternion-angle error over 45 s. CF2 denotes the proposed 1.5× rotor-authority simulation model, not stock-hardware performance.

Why the research is promising

- The architecture augments a conventional controller rather than replacing it. The baseline, allocator, force limits, and sensing interface remain explicit, while the learned authority is bounded.
- The main mechanism is identifiable. Online PES decoder adaptation and persistent activity account for most of the improvement; conductance modulation is nearly inactive in the present saturated-activity regime. This narrows the scientific claim and creates a precise next experiment.
- The same residual interface runs in direct Python and Nengo. Ten matched cases agreed to 2.013×10^{-6} relative mean error, which supports implementation consistency but is not a proof of mathematical equivalence.
- Hardstep exposes decoder lag during actuator-limited slews. The output mask prevents catastrophic opposition and restores near parity without hiding the remaining settling-tail limitation.

Results and their limits

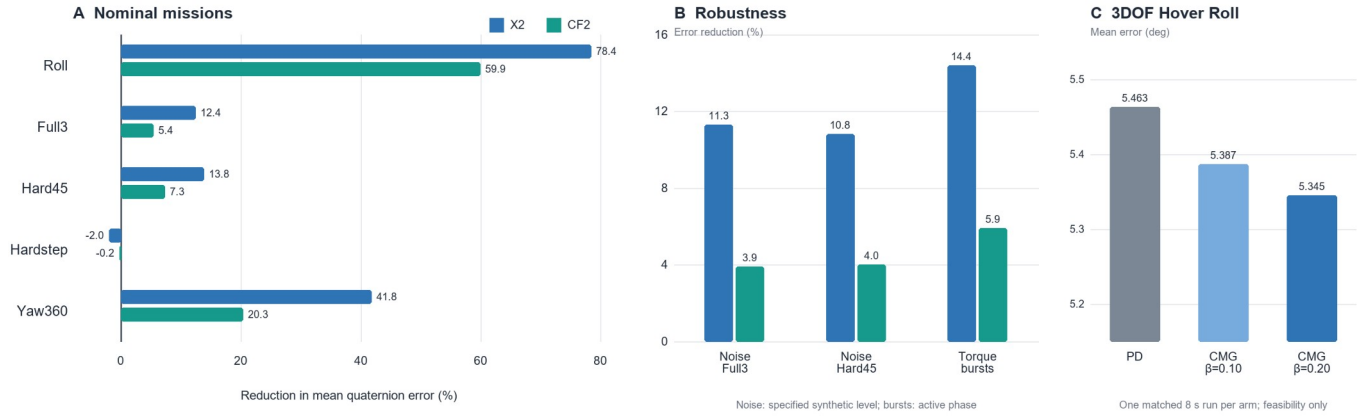


Figure 2 Current evidence. Positive values denote lower error than the matched baseline. Panel C is a single matched physical run per controller and is not a statistical hardware claim.

Hardware hypothesis and decisive validation

The first matched Nengo 3DOF Hover Roll trial reduced mean attitude error from 5.463° with PD to 5.345° with CMG-PES at β = 0.20, a 2.16% improvement, and the signed Roll pair carried the neural activity. This supports real-time feasibility. It does not yet establish repeatable performance because each arm has one matched eight-second run.

The hypothesis is confirmed only if interleaved baseline and CMG trials show a consistent paired improvement across repeated Roll trials, then Full3 and disturbance-recovery tests, without increased aborts, saturation, peak error, or deadline misses. An initial minimum is five paired trials per condition with frozen parameters and reported confidence intervals, followed by a power-based sample-size decision.

Publication path

Venue	Best version of the work	Evidence needed
ACC 2027 [1]	Current control architecture, bounds, ablations, and two-model study	Ready for advisor review; retain the honest Hardstep and hardware limits
IEEE CCTA 2027 [2]	Implementation-focused extension	Repeated 3DOF Hover and preferably QDrone results
IEEE L-CSS [3]	Concise control contribution with rapid journal review	Sharper theorem or stability result and a self-contained six-page treatment
IEEE TCST [4]	Full archival control-technology paper	Multi-mission hardware, repeatability, timing, safety, and matched adaptive baselines
NCE [5] or RA-L [6]	Neuromorphic mechanism or robotics implementation	Broader neuron study plus neuromorphic hardware/energy data, or stronger robot experiments

Venue sources [1] American Control Conference overview | [2] IEEE CCTA 2027 scope | [3] IEEE Control Systems Letters scope | [4] IEEE Transactions on Control Systems Technology scope | [5] Neuromorphic Computing and Engineering scope | [6] IEEE Robotics and Automation Letters scope